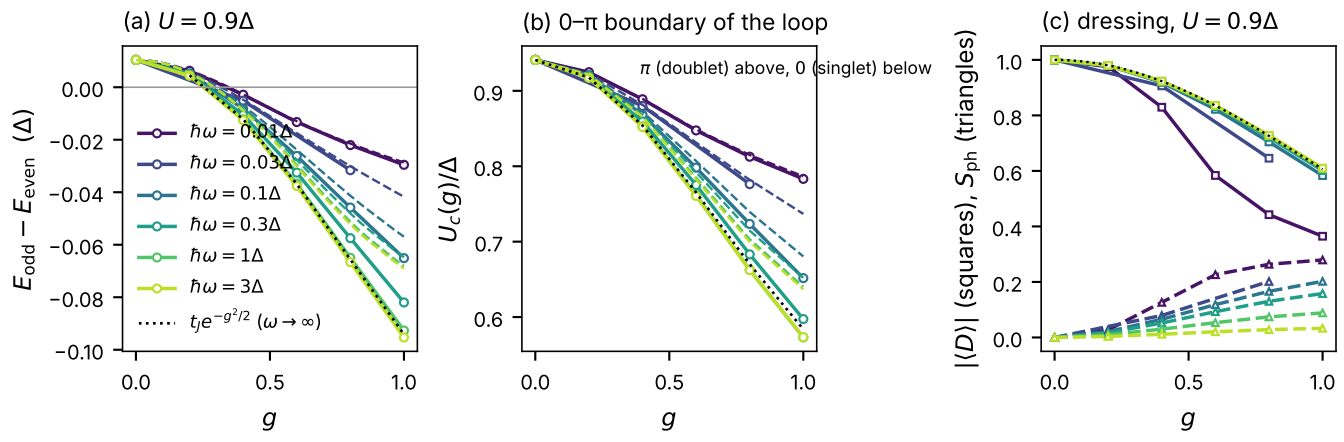
(a) $U = 0.9\Delta$
$E_{\mathrm{odd}} - E_{\mathrm{even}}$ $(\Delta)$
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
$\hbar\omega = 0.01\Delta$
$\hbar\omega = 0.03\Delta$
$\hbar\omega = 0.1\Delta$
$\hbar\omega = 0.3\Delta$
$\hbar\omega = 1\Delta$
$\hbar\omega = 3\Delta$
$t_J e^{-g^2/2}$ $(\omega \to \infty)$
0.0
0.5
1.0
$g$
(b) 0–π boundary of the loop
$\pi$ (doublet) above, 0 (singlet) below
$U_c(g)/\Delta$
0.9
0.8
0.7
0.6
0.0
0.5
1.0
$g$
(c) dressing, $U = 0.9\Delta$
$|\langle D \rangle|$ (squares), $S_{\mathrm{ph}}$ (triangles)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
$g$